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TME–403

B. TECH. (ME) (FOURTH SEMESTER)

MID SEMESTER

EXAMINATION, March, 2024

MANUFACTURING PROCESSES–II

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Draw merchant circle diagram for cutting forces and derive the following components : (CO1)

(i) Write assumptions, chip thickness ratio, velocity relationship, shear plane angle.

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(ii) Expression for cutting, shear and frictional force and relation between them, work done in cutting, shear and frictional.

OR

(b) In an orthogonal cutting operation, the following data have been observed :

Uncut chip thickness = 0.127 mm, Width of cut = 6.35 mm, Cutting speed = 2m/s, Rake angle = 10, Cutting force = 567 N, Thrust force = 227 N, Chip thickness = 0.228 mm.

Determine shear angle, friction angle, shear stress along the shear plane, power and chip velocity. (CO1)

2. (a) What are the operations that can performed on lathe machine ? (CO3)

OR

(b) What are the basic principle and features of planar machine ? (CO3)

3. (a) Discuss tool elements and tool angles of single point cutting tool with neat sketch and give detailed description of its nomenclature. (CO1)

OR

(b) Explain different types of tool wear. (CO1)

4. (a) What are the different materials used for making tools ? What are the basic properties for selecting the tool material ? (CO2)

OR

(b) Explain the thermal aspect of chip formation. What are the factors affecting cutting temperature ? (CO2)

5. (a) Explain dill bit nomenclature with suitable diagram. (CO2)

OR

(b) Explain the different drilling operations. (CO2)

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(ii) Expression for cutting, shear and frictional force and relation between them, work done in cutting, shear and frictional.

OR

(b) In an orthogonal cutting operation, the following data have been observed :

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5. (a) Explain dill bit nomenclature with suitable diagram. (CO2)

OR

(b) Explain the different drilling operations. (CO2)

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